

NAVEEN KUMAR R

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CAREER OBJECTIVE

Motivated BE-EEE undergraduate with hands-on expertise in Verilog HDL, embedded systems (ESP32/Arduino), PCB design, and front-end web development. Seeking a role in VLSI Design / Verification or IT / Software Development where I can apply digital design, programming, and problem-solving skills to build innovative and scalable solutions.

EDUCATION

Qualification	Institution / Board	Year	Score
BE – EEE	PSNACET, Dindigul	2026	CGPA: 8.14 / 10
HSC (Class XII) – CBSE	MONTFORT School, Sivaganga	2022	78%
SSLC (Class X) – CBSE	MONTFORT School, Sivaganga	2020	82%

TECHNICAL SKILLS

HDL / Digital Design: Verilog HDL (NPTEL Certified), Digital Logic, Combinational & Sequential Circuits, FSM

Programming Languages: C, Embedded C, Java (NPTEL Certified), HTML5, CSS3, JavaScript, React.js (Basics), MySQL

EDA & Simulation Tools: Proteus, KiCAD / Eagle, Arduino IDE, Tinkercad, Cadence Virtuoso, Intel ModelSim, Xilinx Vivado

Embedded Platforms: ESP32, Arduino, Sensor Interfacing, Relay Control, UART / I2C / SPI (basics)

Hardware Skills: PCB Design & Fabrication

Industrial / Automation: Siemens PLC Programming, Control Systems, IoT Hardware Integration

Domains: VLSI Digital Design, Embedded Systems, IoT, Front-End Web Development, Automation

PROJECTS

1. Smart Agriculture Monitoring & Automation (ESP32 + IoT)

- Built an ESP32-based embedded system to continuously monitor farm environmental sensors (temperature, humidity, soil moisture).
- Implemented threshold-based logic to automate irrigation and ventilation, reducing manual intervention.
- Integrated Blynk IoT dashboard for real-time remote monitoring and control.
- Skills: Embedded C, sensor interfacing, relay drivers, IoT, hardware-software co-design.

2. Ammonia Detection System for Poultry Farms

- Used MQ-137 gas sensor for accurate ammonia level detection inside poultry enclosures.
- Designed relay-based automatic exhaust fan control triggered on threshold breach.
- Integrated GSM module for instant SMS alerts during unsafe ammonia concentration.
- Skills: Analog sensor interfacing, relay circuits, embedded logic, PCB layout, safety systems.

3. NurtureDose – Responsive Front-End Website

- Developed a fully responsive website using HTML5, CSS3, and JavaScript as part of Infosys-funded UI Training (ICT Academy).
- Designed structured layouts with smooth navigation and cross-device compatibility.
- Skills: HTML, CSS, JavaScript, responsive design, UI/UX fundamentals.

4. Power Leakage Reduction in SRAM Using SK-LCT Based Sense Amplifier

- Designed and analyzed a low-power SRAM architecture using the SK-LCT (Sleepy Keeper Leakage Control Transistor) technique.
- Implemented and simulated SRAM cells and sense amplifiers using Cadence Virtuoso.
- Evaluated power consumption and propagation delay of various power-gating techniques including Sleep, Stack, LECTOR, and SK-LCT.
- Achieved 57.53% reduction in power consumption and 44.70% reduction in delay compared to a conventional 6T SRAM cell.

INTERNSHIP

Electronics Engineering Intern | Vasantha Advanced Systems, Coimbatore (Jan 2025 – Feb 2025)

- Performed PCB assembly (proto & bulk production) and SMT operations in a production environment.
- Worked on microcontroller programming, functional testing, and rapid prototyping of embedded modules.
- Assembled wiring harnesses in compliance with IPC and IATF quality guidelines.
- Conducted quality inspection of assembled PCBs and supported debugging of faulty hardware modules.
- Gained industry-level exposure to manufacturing workflows, quality standards, and team collaboration.

CERTIFICATIONS

- NPTEL – Digital Design with Verilog – IIT / NPTEL Online Certification
- NPTEL – Programming in Java – IIT / NPTEL Online Certification
- PCB Design & Soldering – Industry Training Certificate
- Siemens PLC Programming – Siemens Certified Training

ACHIEVEMENTS & EXTRA-CURRICULAR

- Treasurer – IEEE Industrial Electronics Society Student Branch Chapter
 - Managed event planning, budgeting, and coordination for technical events.
- 1st Prize – DIY Flashing Circuit Competition, IEEE PES (2024)
 - Awarded Best Circuit Design for a creative embedded circuit implementation.
- 2nd Prize – Paper Presentation, Power Gale '24 National Technical Symposium
 - Recognised for an innovative technical paper in power electronics.

AREAS OF INTEREST

• VLSI Design & Verification	• Verilog HDL / Digital Logic Circuits
• Embedded Systems & Microcontroller Programming	• IT / Software Development (Web & Application)
• IoT Hardware Integration	• Automation & Control Systems

PERSONAL DETAILS

DOB:9/11/2004
Languages Known: Tamil (Native), English (Professional)
Nationality: Indian
Personal Strengths: Quick learner, leadership ability, positive attitude, strong analytical and decision-making skills
Location: Sivaganga, Tamil Nadu, India

DECLARATION

I hereby declare that all the information furnished above is true and correct to the best of my knowledge and belief.